

2019 Year 12 Trial HSC Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen.
- Board-approved calculators may be used.
- A reference sheet with standard formulae and integrals is provided.
- In Questions 11–14, show relevant mathematical reasoning and/or calculations.

Total marks – 70

Section I (Pages 3–5)

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II (Pages 6–11)

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section.

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1. The polynomial $P(x) = x^4 - kx^3 - 2x + 33$ has $(x - 3)$ as a factor.

What is the value of k ?

- (A) $-4\frac{4}{9}$ (B) -4
(C) 4 (D) $4\frac{4}{9}$

2. Which is the correct condition for $y = mx + b$ to be a tangent to $x^2 = 4ay$?

- (A) $am^2 + b = 0$ (B) $am^2 - b = 0$
(C) $am + b = 0$ (D) $am - b = 0$

3. The roots of $3x^3 - 2x^2 + x - 1 = 0$ are α, β and γ .

What is the value of $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$?

- (A) $-\frac{1}{9}$ (B) $-\frac{2}{9}$
(C) 1 (D) $\frac{2}{9}$

4. Evaluate $\lim_{x \rightarrow \infty} \left(\frac{1+x}{2-x} \right)$.

(A) -2

(B) -1

(C) 1

(D) 2

5. What is the correct expression for $\int \frac{dx}{9+4x^2}$?

(A) $\frac{1}{4} \tan^{-1} \left(\frac{2x}{3} \right)$

(B) $\frac{1}{3} \tan^{-1} \left(\frac{2x}{3} \right)$

(C) $\frac{1}{6} \tan^{-1} \left(\frac{2x}{3} \right)$

(D) $\frac{2}{3} \tan^{-1} \left(\frac{2x}{3} \right)$

6. A particle moves in simple harmonic motion so that its velocity, v , is given by $v^2 = 6 - x - x^2$.

Between which two points does it oscillate?

(A) $x = 2$ and $x = -3$

(B) $x = -2$ and $x = 3$

(C) $x = 1$ and $x = 2$

(D) $x = 6$ and $x = 3$

7. $\tan^{-1}(-1) =$

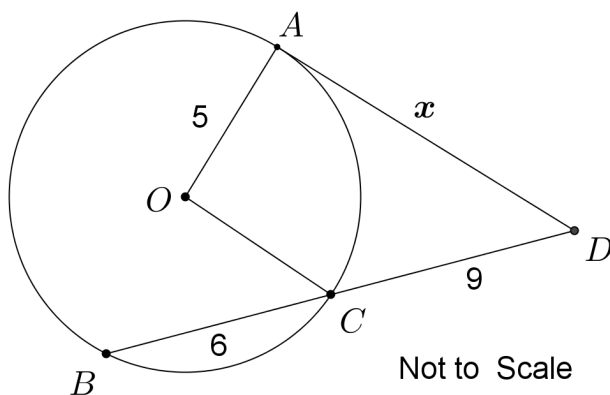
(A) $\frac{-3\pi}{4}$

(B) $\frac{-\pi}{4}$

(C) $\frac{\pi}{4}$

(D) $\frac{3\pi}{64}$

8.



Segment AD lies on a tangent to the circle centre, O , radius 5 cm.

BC is 6 cm and CD is 9 cm.

Find the exact length of AD .

(A) $\sqrt{15}$

(B) $3\sqrt{6}$

(C) $3\sqrt{10}$

(D) $3\sqrt{15}$

9. The general solution for $\cos 2x = -\frac{1}{2}$, where $n = 0, \pm 1, \pm 2, \dots$ is

(A) $x = n\pi + (-1)^n \frac{\pi}{3}$

(B) $x = n\pi + (-1)^n \frac{\pi}{6}$

(C) $x = n\pi \pm \frac{\pi}{6}$

(D) $x = n\pi \pm \frac{\pi}{3}$

10. A flat semi-circular disc is being heated so that the rate of increase of the area ($A \text{ m}^2$),

after t hours, is given by $\frac{dA}{dt} = \frac{1}{4}\pi t$

Initially the disc has a radius of 4 metres.

Which of the following is the correct expression for the area after t hours?

(A) $A = \frac{1}{4}\pi t^2 + 8\pi$

(B) $A = \frac{1}{8}\pi t^2 + 8\pi$

(C) $A = \frac{1}{4}\pi t^2 + 16\pi$

(D) $A = \frac{1}{8}\pi t^2 + 16\pi$

End of Section I

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section .

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks) Use the Question 11 Writing Booklet. **Marks**

a) Calculate the acute angle between the lines $x - 5y - 2 = 0$ and $x - 2y = 0$ to the nearest degree. **2**

b) Solve the inequality $\frac{3x - 2}{x + 1} \geq 5$. **2**

c) i) Express $\cos 2x$ in terms of $\sin^2 x$. **1**

ii) Hence evaluate $\lim_{x \rightarrow 0} \frac{\cos 2x - 1}{x \sin x}$. **1**

d) Evaluate $\int_0^3 x\sqrt{9 - x^2} dx$ using the substitution $u = 9 - x^2$. **3**

e) If A is the point $(-2, -1)$ and B is the point $(1, 5)$, find the coordinates of the point P which divides the interval AB externally in the ratio 2:5. **2**

f) Ms Namvar bought a slurpy in Port Douglas which had a temperature of 5°C . The temperature in Port Douglas was 35°C . The slurpy warms at a rate proportional to the difference between the air temperature and the temperature (T) of the slurpy.

That is, T satisfies the equation $\frac{dT}{dt} = k(T - 35)$.

i) Show that $T = 35 + Ae^{kt}$ satisfies this equation. **1**

ii) If the temperature of the slurpy after ten minutes is 10°C , find its temperature, to the nearest whole degree, after 20 minutes. **3**

End of Question 11

Question 12 (15 Marks) Use the Question 12 Writing Booklet.

Marks

- a) The polynomial $P(x)$ is given by $P(x) = x^3 + bx^2 + cx - 10$ where b and c are constants. The three zeroes of $P(x)$ are -1 , 2 and α .
- i) Find the values of b and c . 2
- ii) Hence or otherwise find the value of α . 1
- b) The equation $2x^3 - 6x + 1 = 0$ has roots α, β and γ . 2
- Evaluate $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.
- c) By considering the expansion for $\tan(\alpha - \beta)$, find x so that 3
- $$\tan^{-1} x = \tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{3}\right).$$
- d) Consider the function $f(x) = e^{2x} + 6e^x + 9$.
- i) Explain why $y = f(x)$ has an inverse function $y = f^{-1}(x)$ for all x . 1
- ii) Draw a neat sketch of $y = f(x)$ and $y = f^{-1}(x)$, 2
- showing all intercepts and asymptotes.
- iii) Find the equation of the inverse function in terms of x . 3
- iv) Hence or otherwise solve $e^{2x} + e^x = 6$. 1

End of Question 12

Question 13 (15 Marks) Use the Question 13 Writing Booklet.

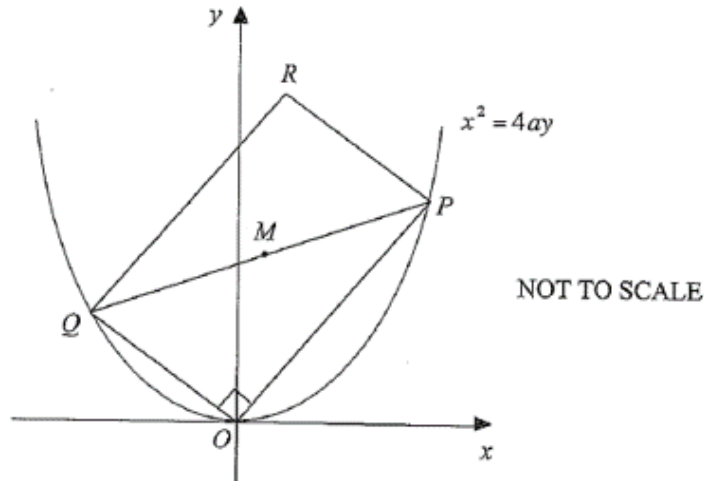
Marks

- a) Evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^2 2x \, dx$. **3**
- b) Given the parametric equations in terms of θ ,
 $x = 3 \sin \theta$ and $y = 4 \cos \theta$,
find the Cartesian equation. **1**
- c) i) Express $3 \sin x - 2 \cos x$ in the form $R \sin(x - \alpha)$. **2**
- ii) Hence solve $3 \sin x - 2 \cos x = 1$, $0 \leq x \leq \frac{\pi}{2}$. **1**
Give your answer correct to 3 significant figures.

Question 13 continued on next page

Question 13 (continued)

- d) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. The chords OP and OQ meet at right angles at the origin. M is the midpoint of the chord PQ . R is a point (not on the parabola) such that $OPRQ$ is a rectangle, as shown in the diagram below.



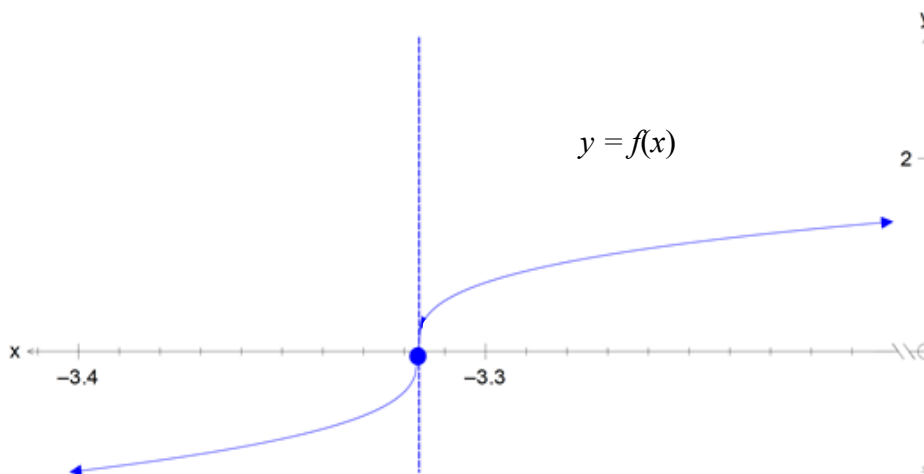
- | | |
|---|---|
| i) Show that $pq = -4$. | 1 |
| ii) Explain why R has the coordinates $(2a(p + q), a(p^2 + q^2))$. | 1 |
| iii) Find the equation of the locus of R . | 2 |
- e) A particle oscillates in a straight line under simple harmonic motion. At time, t , it has displacement of x metres from a fixed point O on the line. It's velocity, $v \text{ ms}^{-1}$, is given by $v^2 = 32 + 8x - 4x^2$.
- | | |
|---|---|
| i) Find an expression for the particle's acceleration in terms of x . | 1 |
| ii) Find the period and amplitude of the motion. | 2 |
| iii) Find the maximum speed of the particle. | 1 |

End of Question 13

Question 14 (15 Marks) Use the Question 14 Writing Booklet.

Marks

- a) Prove by mathematical induction that $(3^{2n} - 1)$ is divisible by 8, for all integers $n = 1, 2, 3, \dots$ **3**
- b) A point $P(x, y)$ moves so that its distance from $A(8, -2)$ is equal to twice its distance from $B(-1, 4)$. Find its locus in algebraic form and describe the locus geometrically. **3**
- c) The curve $f(x) = (x^3 - 12x)^{\frac{1}{3}}$ is shown below.

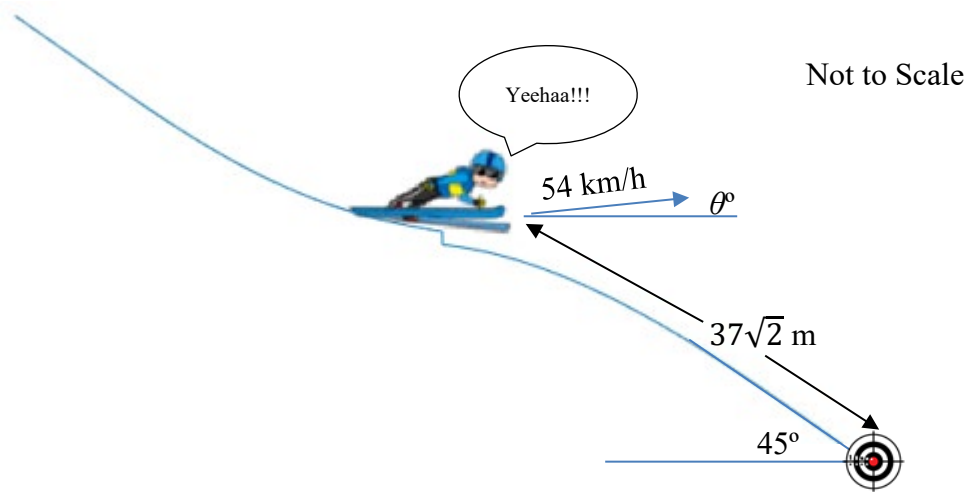


- i) It can be seen that $y = f(x)$ crosses the x -axis near $x = -3.3$. Use one application of Newton's method to obtain another approximation to the root of $f(x) = 0$. **2**
- ii) Explain why using $x = -3.3$ as a first approximation does not produce a better approximation to the root than the original approximation. **1**

Question 14 continued on next page

Question 14 (continued)

- d) Mr Laurendet has taken some time off work to organize a ski-jump training session for the upcoming snow excursion. He has calculated that if he skis down a slope at Perisher at 54 km/h and launches of a jump he can land on a target $37\sqrt{2}$ metres down the slope. It is known that the slope below the jump falls away at an average rate of 45° .



Let acceleration due to gravity to be $g = -10 \text{ ms}^{-2}$ and Mr Laurendet's angle of projection above the horizontal to be θ ,

- i) Show that his trajectory path is given by the equation 3

$$y = -\frac{x^2 \sec^2 \theta}{45} + x \tan \theta.$$

- ii) Hence, find the smallest positive angle of projection, θ , to the nearest whole degree, that enables him to land on his target $37\sqrt{2}$ m away. 3

End of Question 14

End of Examination

2019 MATHEMATICS EXT 1

SOLUTIONS

- 1) C 2) A 3) D 4) B 5) C
6) A 7) B 8) D 9) D 10) B

Question 11

$$\begin{aligned} a) m_1 &= \frac{1}{5} \quad m_2 = \frac{1}{2} \quad \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{\frac{1}{5} - \frac{1}{2}}{1 + \frac{1}{5} \times \frac{1}{2}} \right| \\ &= \frac{3}{11} \\ \theta &\doteq 15^\circ \end{aligned}$$

b) critical pts $x \neq -1$ $3x-2 = 5(x+1)$
 $x = -\frac{7}{2}$

When $x = -2$ LHS = 8, True
 $x = 0$ LHS = -2, False
 $x = -10$ LHS = $-\frac{32}{9}$, False
 $\therefore -\frac{7}{2} \leq x < -1$

c) i) $\cos 2x = 1 - 2\sin^2 x$

ii) $\lim_{x \rightarrow 0} \frac{1 - 2\sin^2 x - 1}{x \sin x} = \lim_{x \rightarrow 0} \frac{-2\sin x}{x} = -2$

d) $u = 9 - x^2$
 $du = -2x dx$
 $x=0 \quad u=9$
 $x=3 \quad u=0$
 $\int_0^3 \sqrt{9-x^2} dx = -\frac{1}{2} \int_9^0 u^{\frac{1}{2}} du$
 $= \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_0^9$
 $= \frac{1}{3} (27 - 0)$
 $= 9$

e) $\frac{m x_2 - n x_1}{m - n} = \frac{2 \times 1 + 5 \times 2}{2 - 5} = \frac{10 + 5}{-3} = -5$
 $\frac{m y_2 - n y_1}{m - n} = \frac{10 + 5}{-3} = -5$

$P(-4, -5)$

f) i) $T = 35 + A e^{kt}$ $A e^{kt} = T - 35$

$\frac{dT}{dt} = k A e^{kt}$
 $= k(T - 35)$ as required.

ii) $t=0 \quad 5 = 35 + A e^0 \therefore A = -30$

$t=10 \quad 10 = 35 - 30 e^{10k}$

$e^{10k} = \frac{25}{30}$

$k = \frac{\ln \frac{5}{6}}{10}$

$T = 35 - 30 e^{\frac{\ln \frac{5}{6}}{10} t}$ $t=20 \quad T = 14^\circ$

Question 12

a) $P(-1) = -1 + b - c - 10 = 0$ $P(2) = 8 + 4b + 2c - 10 = 0$

$b - c = 11$ - (1) $4b + 2c = 2$
 $2b + c = 1$ - (2)

(1) - (2) $\Rightarrow 3b = 12, b = 4, c = -7$

ii) $-1 \times 2 \times \alpha = 10$ or $-1 + 2 + \alpha = -4$
 $\alpha = -5$ $\alpha = -5$

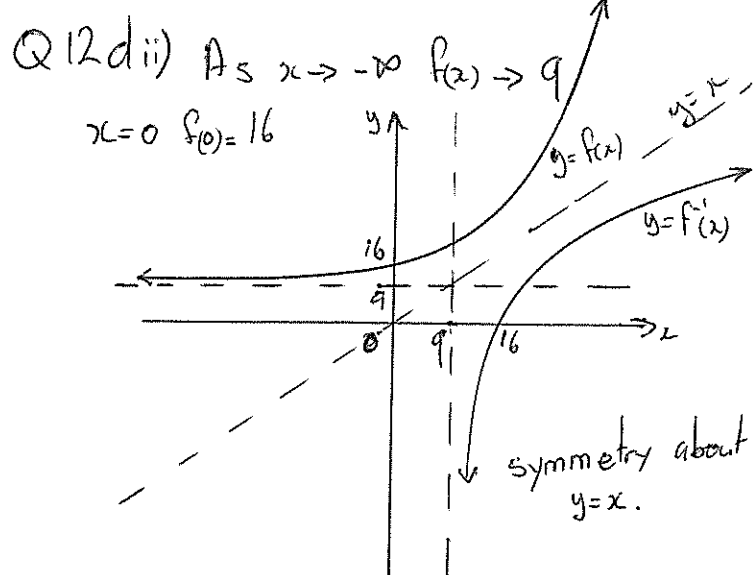
b) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$

$\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{6}{2} = -3$ $\alpha\beta\gamma = -\frac{1}{2} \times \frac{1}{\beta} + \frac{1}{\gamma} = 6$

c) let $\tan^{-1} \frac{1}{2} = \alpha$ and $\tan^{-1} \frac{1}{3} = \beta$

$\tan [\tan^{-1} x] = \tan [\alpha - \beta]$
 $x = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$
 $= \frac{\frac{1}{2} - \frac{1}{3}}{1 + \frac{1}{2} \times \frac{1}{3}} = \frac{1}{7}$

d) i) $y = e^{2x}$ and $y = e^x$ are monotonic increasing functions, so $f(x)$ is a monotonic increasing function for all x . Hence a horizontal line will cross only once, so $f^{-1}(x)$ exists for all x .



iv)

$$x = e^{2y} + 6e^y + 9$$

$$\text{let } m = e^y \quad e^{2y} = m^2$$

$$x = m^2 + 6m + 9$$

$$= (m+3)^2$$

$$m+3 = \pm\sqrt{x} \quad \text{as } m > 0 \quad m+3 = \sqrt{x}$$

$$e^y + 3 = \sqrt{x}$$

$$e^y = \sqrt{x} - 3$$

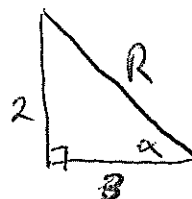
$$y = \ln(\sqrt{x} - 3)$$

c) i) $3 \sin x - 2 \cos x$
 $R \sin x \cos \alpha - R \cos x \sin \alpha$

$$\cos \alpha = \frac{3}{R} \quad \sin \alpha = \frac{2}{R}$$

$$R = \sqrt{2^2 + 3^2}$$

$$= \sqrt{13} \quad \tan \alpha = \frac{2}{3}$$



$$3 \sin x - 2 \cos x = \sqrt{13} \sin(x - \tan^{-1} \frac{2}{3})$$

$$\alpha \doteq 0.588$$

ii) $\sqrt{13} \sin(x - 0.588) = 1$

$$\sin(x - 0.588) = \frac{1}{\sqrt{13}}$$

$$x - 0.588 = 0.281$$

$$x \doteq 0.869 \quad (3 \text{ sig fig})$$

d) i) $m_{op} \times m_{oa} = -1$

$$m_{op} = \frac{ap^2}{2ap} \quad m_{oa} = \frac{aq^2}{2aq}$$

$$= \frac{p}{2} \quad = \frac{q}{2}$$

$$\therefore \frac{p}{2} \times \frac{q}{2} = -1$$

$$pq = -4 \quad \text{as required}$$

ii) $O(0,0) \rightarrow Q$

add $2aq$ to x coord. $(0+2aq)$

add aq^2 to y coord. $(0+aq^2)$

$$P(2ap, ap^2) \rightarrow R$$

add $2aq$ to x coord $(2ap+2aq)$

add aq^2 to y coord (ap^2+aq^2)

$$\therefore R[2a(p+q), a(p^2+q^2)]$$

$$x = 2a(p+q) \quad y = a(p^2+q^2)$$

$$p+q = \frac{x}{2a}$$

$$p^2+q^2 = \frac{y}{a}$$

$$(p+q)^2 - 2pq = \frac{y}{a}$$

$$\left(\frac{x}{2a}\right)^2 - 2x(4) = \frac{y}{a}$$

$$\frac{x^2}{4a^2} + 8 = \frac{y}{a}$$

$$x^2 = 4ay - 32a^2$$

Question 13

a) $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^2 2x \, dx = \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos 4x + 1 \, dx$

$$= \frac{1}{2} \left[\frac{\sin 4x}{4} + x \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= \frac{1}{8} \left[\sin 2\pi - \sin \frac{4\pi}{3} \right] + \frac{1}{2} \left[\frac{\pi}{2} - \frac{\pi}{3} \right]$$

$$= \frac{\sqrt{3}}{16} + \frac{\pi}{12}$$

b) $\sin \theta = \frac{x}{3}$

$$\cos \theta = \frac{y}{4}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\frac{x^2}{9} + \frac{y^2}{16} = 1$$

Q13 a) $v^2 = 32 + 8x - 4x^2$

$$\frac{1}{2}v^2 = 16 + 4x - 2x^2$$

$$\frac{d(\frac{1}{2}v^2)}{dx} = 4 - 4x$$

$$= -4(x-1)$$

ii) $v=0$ $4x^2 - 8x - 32 = 0$
 $2x^2 - 4x - 16 = 0$
 $(2x-8)(x+2) = 0$
 $x = -2$ or $x = 4$

$\therefore$ oscillates between $x = -2, 4$
 ie amplitude = 3

$$T = \frac{2\pi}{n} \quad n^2 = 4 \therefore n = 2$$

$$= \pi \text{ seconds}$$

iii) Max speed at centre of oscillation

$$x=1 \quad v^2 = 32 + 8 - 4$$

$$= 36$$

$$\therefore \text{max speed} = \underline{6 \text{ m/s}}$$

Q14 $n=1 \quad 3^{2n} - 1 = 8 \therefore \text{true for } n=1$

Assume true for $n=k$

$$3^{2k} - 1 = 8M \quad (M \text{ is integer valued})$$

Prove true for $n=k+1$

$$3^{2(k+1)} - 1 = 3^2 \cdot 3^{2k} - 1$$

$$= 9(8M+1) - 1$$

$$= 72M + 8$$

$$= 8(9M+1) \therefore \text{divisible by 8}$$

$\therefore$ If true for $n=k$, then true for $n=k+1$

As it is true for $n=1$, then by the principle of mathematical induction, it is true for all integers $n=1, 2, 3, \dots$

b) $AP = 2BP$

$$AP^2 = 4BP^2$$

$$(x-8)^2 + (y+2)^2 = 4[(x+1)^2 + (y-4)^2]$$

$$x^2 - 16x + 64 + y^2 + 4y + 4 = 4x^2 + 8x + 4 + 4y^2 - 32y + 64$$

$$3x^2 + 24x + 3y^2 - 36y = 0$$

$$x^2 + 8x + y^2 - 12y = 0$$

$$(x+4)^2 + (y-6)^2 = 52$$

circle centre $(-4, 6)$ radius $= \sqrt{52}$
 $= \underline{2\sqrt{13}}$

c) i) $f(x) = (x^3 - 12x)^{\frac{1}{3}}$

$$f'(x) = \frac{1}{3}(3x^2 - 12)(x^3 - 12x)^{-\frac{2}{3}}$$

$$= \frac{x^2 - 4}{(x^3 - 12x)^{\frac{2}{3}}}$$

$$x_1 = -3.3 \quad f(-3.3) \doteq 1.542$$

$$f'(-3.3) \doteq 2.900$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= -3.3 - \frac{1.542}{2.9}$$

$$= \underline{-3.83} \quad (3 \text{ sig fig})$$

ii) Newton's method uses the x -intercepts of tangents to find an approximate.

At $x = -3.3$ the slope of $y = f(x)$ is not very steep and this pushes the tangent away from the root rather than closer.

Q14d

i) $\dot{x} = V \cos \alpha$ $\dot{y} = -gt + V \sin \alpha$

$V = 54 \text{ km/h} = 15 \text{ ms}^{-1}$

$\dot{x} = 15 \cos \theta$ $\dot{y} = -10t + 15 \sin \theta$

$x = 15t \cos \theta + c$ $y = -5t^2 + 15t \sin \theta + c$

$t = 0$ $x = 0$ $y = 0$ $\therefore$ Both $c = 0$.

$x = 15t \cos \theta$ $y = -5t^2 + 15t \sin \theta$

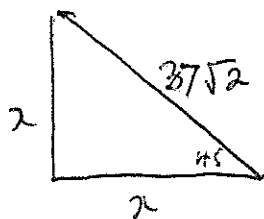
$t = \frac{x}{15 \cos \theta}$

$y = -5 \left(\frac{x}{15 \cos \theta} \right)^2 + 15 \left(\frac{x}{15 \cos \theta} \right) \sin \theta$

$= \frac{-x^2}{45 \cos^2 \theta} + \frac{x \sin \theta}{\cos \theta}$

$= -\frac{x^2 \sec^2 \theta}{45} + x \tan \theta.$

ii) $x = 37$ $y = -37$



$x^2 + x^2 = (37\sqrt{2})^2$

$2x^2 = 37^2 \times 2$

$x = 37, y = 37$

$-37 = -\frac{(37^2) \sec^2 \theta}{45} + 37 \tan \theta$

$\sec^2 \theta = 1 + \tan^2 \theta$

$-37 = -\frac{37^2}{45} (1 + \tan^2 \theta) + 37 \tan \theta$

$-45 = -37 - 37 \tan^2 \theta + 45 \tan \theta$

$37 \tan^2 \theta - 45 \tan \theta - 8 = 0$

let $m = \tan \theta$

$37m^2 - 45m - 8 = 0$

$m = \frac{45 \pm \sqrt{45^2 + 4 \times 37 \times 8}}{74}$

$= 1.373 \text{ or } -0.157$

as $m > 0$

$\theta = \tan^{-1}(1.373)$
 $= \underline{\underline{54^\circ}}$